

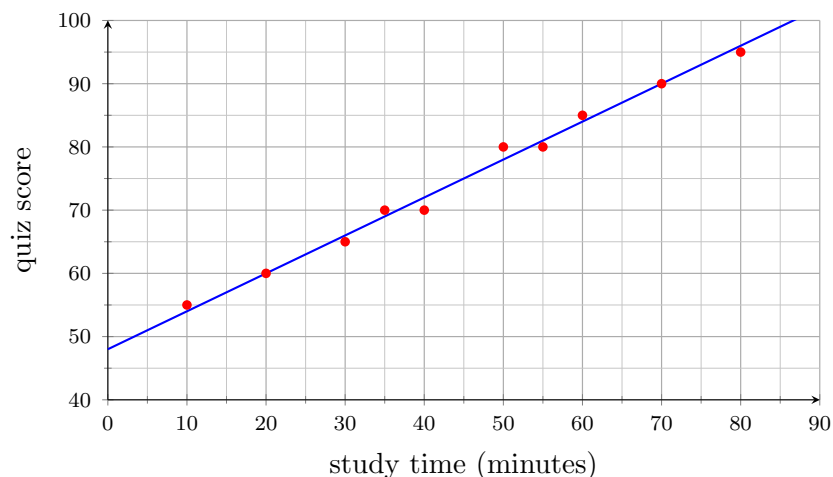
Making and Qualifying Predictions from Scatter Plots

Reading a scatter plot, predicting with its trend line, and judging when a prediction can be trusted

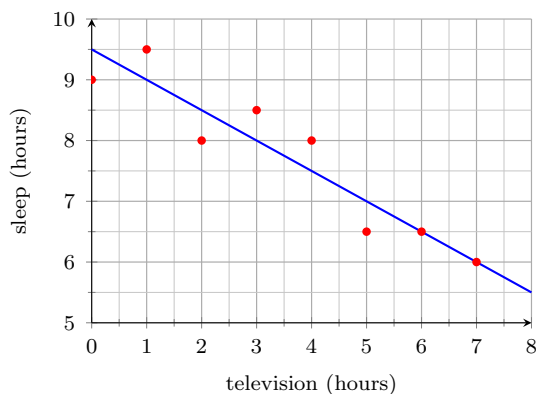
Directions

- The three displays below are used by every problem. Each problem names the display it needs.
- Where a trend line is given as an equation, use the equation, not a ruler.
- Every prediction has to be judged as well as calculated: say when a number the line produces is one you would actually rely on.

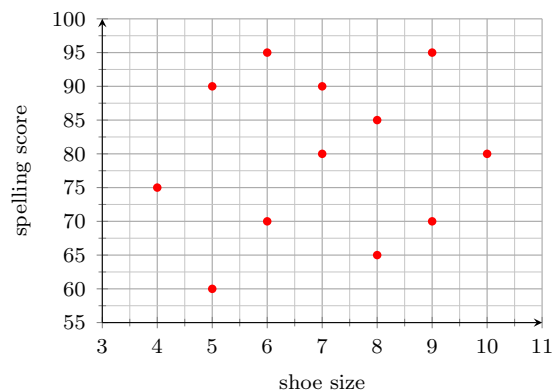
Display A — study time and quiz score for ten students. Trend line: $y = 0.6x + 48$.



Display B — daily television and sleep for eight students. Trend line: $y = 9.5 - 0.5x$.



Display C — shoe size and spelling score for twelve students. No trend line is drawn.



1. Display A. Any prediction you make from this display rests on the students who were actually measured. How many students are plotted?

Answer: _____

2. Display A. One student studied for 35 minutes. Read that student's actual quiz score off the plot — you will compare it with a prediction later.

Answer: _____

3. Display A. Use the trend line $y = 0.6x + 48$ to predict the quiz score of a student who studies for 45 minutes.

Answer: _____

4. Display B. Use the trend line $y = 9.5 - 0.5x$ to predict the sleep of a student who watches 5.5 hours of television a day.

Answer: _____ hours

5. Display A. A parent asks what score 200 minutes of study would produce.

(a) Work out what the trend line predicts for $x = 200$.

Answer: _____

(b) Give two reasons this prediction should not be trusted.

Answer: _____

6. Display B.

(a) How much sleep did the student who watched 6 hours of television get?

Answer: _____ hours

(b) Describe the direction of the association, and say what that direction means for any prediction you make from this display.

Answer: _____

7. Display A. One student studied for 60 minutes and scored 85.

(a) What score does the trend line predict for 60 minutes?

Answer: _____

(b) By how much does the actual score exceed the prediction?

Answer: _____

8. Display B. A prediction is easier to judge if you know what a typical value looks like. Find the mean sleep time of the eight students.

Answer: _____ hours

9. Display C.

(a) What is the lowest spelling score in the display?

Answer: _____

(b) A classmate wants to predict a student's spelling score from their shoe size. Explain whether this display supports such a prediction.

Answer: _____

10. Display A. A student wants to score 96 on the next quiz. Using the trend line $y = 0.6x + 48$, how many minutes of study does the model call for?

Answer: _____ min

11. Display B. Somebody claims to watch 20 hours of television a day.

(a) What does the trend line $y = 9.5 - 0.5x$ predict for their sleep?

Answer: _____ hours

(b) Explain why that number cannot be treated as a real prediction.

Answer: _____

12. Display A. A school newsletter prints the headline “Studying causes higher quiz scores” beside this display.

- (a) Use the trend line to predict the score of a student who studies 25 minutes.

Answer: _____

- (b) Two of the plotted students studied 30 and 35 minutes and scored 65 and 70. Find the difference between those two scores.

Answer: _____

- (c) Explain whether Display A justifies the headline, and give one other explanation that would fit the same points.

Answer: _____